Comparative Analysis of Quantum Support Vector Classifiers (QSVC) and Variational Quantum Classifier (QVC) for Machine Learning Tasks

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Final Report

Ethan Hammond

Student ID: 19314353

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Supervisor: Professor Simon Caton



UCD School of Computer Science
University College Dublin

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Abstract

Quantum Computing has garnered significant attention in recent years due to its potential applications in various fields of Computer Science, such as cryptography, chemistry, Big Data, machine learning, optimization, the Internet of Things, Blockchain, and communication. As a result, there has been a growing interest in combining Quantum Information Processing with machine learning to create a new field known as Quantum Machine Learning. This field aims to improve upon and solve many challenges faced by classical machine learning techniques, such as energy and time consumption, and kernel estimation.

The goal of this paper is to present and summarize a range of state-of-the-art classification models within the field of Quantum Machine Learning, and to provide a comparison of simulated Quantum models and their classical counterparts with actual Quantum devices. To accomplish this, we propose to run these models on an IBM Quantum device in order to assess their performance and compare the results with simulated Quantum machines and classical algorithms.

Link to the Code Repository: https://gitlab.com/ethanzebedee/final-year-project

Link to a video demonstration of VQC deployment: https://youtu.be/s_GrEdc55i8

Link to a video demonstration of QSVC deployment: https://youtu.be/pmZLpZ0vmUc

Chapter 1: Project Specification

Project Mentor: Dr. Simon Caton

Subject: Quantum Machine Learning

Project Type: Quantum and Machine Learning

Hardware Requirements:

- Quantum computer or quantum emulator, such as IBM Quantum or Rigetti Quantum Computing, for running and testing the Quantum Machine Learning algorithms.
- A classical computer with enough processing power and memory to run our classical preprocessing and postprocessing tasks.

Software Requirements:

- Python version 3.8.8 or higher for programmatic implementation of the VQC and QSVC algorithms, and their classical counterparts.
- Jupyter Notebooks version 6.3.0 or higher for implementing and analyzing the algorithms using Python.
- Quantum computing libraries and frameworks, such as Qiskit or PennyLane, for implementing and running quantum circuits and algorithms.
- Data sets for testing and evaluating the performance of the VQC and QSVC algorithms.
- Software for visualizing and analyzing the results of the algorithms, such as Matplotlib or Seaborn.
- Libraries for implementing basic hybrid quantum-classical machine learning algorithms such as Tensorflow Quantum.

Preassigned: No

In this project, we aim to explore the potential of quantum computers for machine learning tasks. We will focus on comparing the performance of classical machine learning algorithms with simple quantum machine learning models. The goal is to devise ways to benchmark quantum machine learning classification models and to understand how they compare to classical algorithms in terms of cost, performance, speed, interpretability, reproducibility, and potential for raising venture capital. Our objective is to build upon the findings of previous research [1] and investigate whether the results and observations generated in simulation can be replicated on a real quantum device. We will focus on supervised learning tasks using quantum computers. Overall, this project aims to contribute to the growing field of quantum machine learning by exploring the potential of quantum computers for machine learning tasks and comparing their performance to classical algorithms. The results of this project will be relevant to a range of applications in artificial intelligence and machine learning, and could have significant implications for the development of quantum machine learning models and their use in practice. Thus the goals of this project are as follows:

1.1 Problem Statement:

The goal of this project is to implement and compare the performance of both a variational quantum classifier (VQC) algorithm and a quantum support vector circuit (QSVC) for various classification tasks as against their classical equivalent algorithms. The focus of this project is to explore the potential advantages of using quantum machine learning algorithms for these tasks, how their performance can vary between simulations and real devices, and to determine the extent to which they can outperform classical machine learning algorithms in terms of accuracy and runtime.

1.2 Objectives

The main objectives of this project are as follows:

- Implement a VQC algorithm following the approach outlined in [2, 3].
- Implement a QSVC algorithm as described in [4, 5].
- Evaluate the performance of the VQC and QSVC algorithms on various datasets with different characteristics, such as the number of classes, number of features, and data types.
- Evaluate the performance of the VQC and QSVC algorithms on both a simulated Quantum machine and a real Quantum device.
- Compare the results of the VQC and QSVC algorithms to those obtained using classical machine learning algorithms in terms of accuracy and runtime.
- Compare the performance of the VQC and QSVC algorithms when performed in simulation and when performed on a real quantum device.
- Determine the potential advantages and limitations of using quantum machine learning algorithms for these tasks.

1.3 Methodology

To achieve the objectives of this project, we will follow the following methodology:

- Select a suitable dataset for testing the performance of the VQC and QSVC algorithms.
- Preprocess the dataset to ensure that it is in a suitable format for input into the quantum computer. This may involve converting the data into a numerical format and scaling it to a specific range.
- Implement the VQC and QSVC algorithms using quantum circuits and quantum gates to perform the necessary calculations and manipulations.
- Train and evaluate the performance of the VQC and QSVC algorithms on the selected dataset.

- Compare the results of the VQC and QSVC algorithms to those obtained using classical machine learning algorithms. Analyze the results and determine the potential advantages and limitations of using quantum machine learning algorithms for these tasks.

Chapter 2: **Introduction**

In this project, we will delve into quantum machine learning and explore the vast potential of quantum computers for machine learning tasks. Quantum machine learning is a relatively new field that combines quantum computing with machine learning, leveraging the unique features of quantum computers to enable new algorithms and approaches for machine learning tasks. In recent years, there has been a surge of interest in quantum machine learning, as researchers have explored the possibility of quantum computers for supervised and unsupervised learning tasks, as well as the training of neural networks.

One of the key areas of focus in quantum machine learning is the development of algorithms that can harness the power of quantum computers to perform machine learning tasks more efficiently than classical algorithms. This includes exploring the benefits of quantum computers for tasks such as kernel estimation and time and energy consumption, as well as the use of quantum computers for reinforcement learning. In addition, researchers have been working to characterize the quantum information processing capabilities of various quantum machine learning architectures and to understand how quantum computers can be used to improve or optimize machine learning performance.

In this project, we aim to further build upon the growing field of quantum machine learning and contribute to the current exploration of the potential of using quantum computers for machine learning tasks and comparing their performance to classical algorithms. Our goal is to devise ways to compare quantum machine learning models with their classical equivalents in order to understand how they compare to classical algorithms in terms of cost, performance, speed, and interpretability. To achieve this, we will build upon the findings of previous research and investigate whether the results and observations generated in simulation can be replicated on a real quantum device.

To fully evaluate the capabilities of quantum machine learning, we will focus on a range of supervised learning tasks, as well as the training of neural networks using quantum computers. We will run a number of experiments to gauge the performance of classical machine learning algorithms with quantum machine learning algorithms run in simulation and on a real quantum device. By conducting a thorough analysis of these various tasks and approaches, we hope to gain a deep understanding of what quantum computers can do in the field of machine learning, the disparity between simulated quantum performance and real quantum performance, and to identify the areas in which they show the most promise.

We believe that this project is worthwhile because quantum computers could significantly benefit machine learning by optimising algorithms by several orders of magnitude and enabling new types of machine learning algorithms that are not possible with classical computers. In addition, the field of quantum machine learning is rapidly growing and there is a great deal of excitement and interest in its applications. By exploring the capabilities of quantum computers for machine learning tasks and comparing their performance to classical algorithms, we can gain valuable insights that will shape the future of this exciting field.

In order to achieve our objectives and successfully complete this project, we have developed a detailed plan that outlines the various steps we will take. This plan includes conducting a literature review to familiarize ourselves with the current state of the field, selecting appropriate algorithms and tasks to test on quantum computers, setting up a development environment and preparing the algorithms for testing, and running the algorithms on a real quantum device. In addition, we

will analyze the results of our experiments and write a comprehensive report outlining our findings and conclusions.

In summary, this project represents a valuable opportunity to contribute to the growing field of quantum machine learning and to gain a deeper understanding of the potential of quantum computers for machine learning tasks. We are confident that with careful planning and a thorough approach outlined in this paper we can successfully complete this project and make a meaningful contribution to this exciting and rapidly evolving field of research.

Chapter 3: Related Work and Ideas

The field of quantum machine learning has gained significant attention in recent years due to its potential to transform traditional machine learning techniques. As a rapidly growing area at the intersection of quantum computing and machine learning, it leverages the unique features of quantum computers to enable new algorithms and approaches for machine learning tasks.

In this literature review, we will delve into the research we have conducted in the field of Quantum Machine Learning and Classification problems within that field. We will provide a short explanation of quantum computing (see Section 3.1) and quantum machine learning (see Section 3.2). We then explore the motivations for research into this field (see Section 3.3) and compare the performance of quantum and classical models (see Section 3.4), including the current state-of-the-art in quantum machine learning (see Section 3.5). We will also explore the use of simulation and actual quantum devices in the development and evaluation of these models (see Section 3.6). Finally, we will discuss the application of quantum machine learning to classification problems (see Section 3.7), including the various approaches that have been proposed and the challenges that need to be overcome (see Section 3.8). By providing a comprehensive overview of the current state of the field, we aim to highlight the potential of quantum machine learning for advancing artificial intelligence and machine learning.

3.1 Quantum Computing

Quantum computing is a rapidly developing field that has the potential to revolutionise many areas of science and technology. At its core, quantum computing is a new way of processing and storing information that exploits the strange and counter-intuitive properties of quantum mechanics [6]. Quantum computers operate using quantum bits, or qubits, which are quantum mechanical systems that can represent both a 0 and a 1 simultaneously [7]. This property, known as superposition, allows quantum computers to solve certain problems and calculations much faster than classical computers, which can only perform calculations one bit at a time. [8]

In addition to superposition, quantum computers can also make use of another quantum mechanical phenomenon called entanglement [7]. Entanglement allows the state of one qubit to be correlated with the state of another, even if they are separated by large distances [6]. This allows quantum computers to perform certain types of parallel calculations that are not possible on classical computers [7].

Quantum computers operate using quantum gates, which manipulate the state of qubits in a quantum circuit. Quantum algorithms, such as Shor's algorithm for factoring large numbers, can then be implemented using these quantum gates[10]

While quantum computers are still in the early stages of development, they have already shown promise in a number of areas, including optimization problems, machine learning, and cryptography [6]. However, building and maintaining quantum computers is still a challenging task, due to the delicate and fragile nature of quantum systems [7]. As a result, much research is currently being conducted to improve the stability and reliability of quantum computers, as well as to develop new algorithms and applications that can take advantage of their unique capabilities [6].

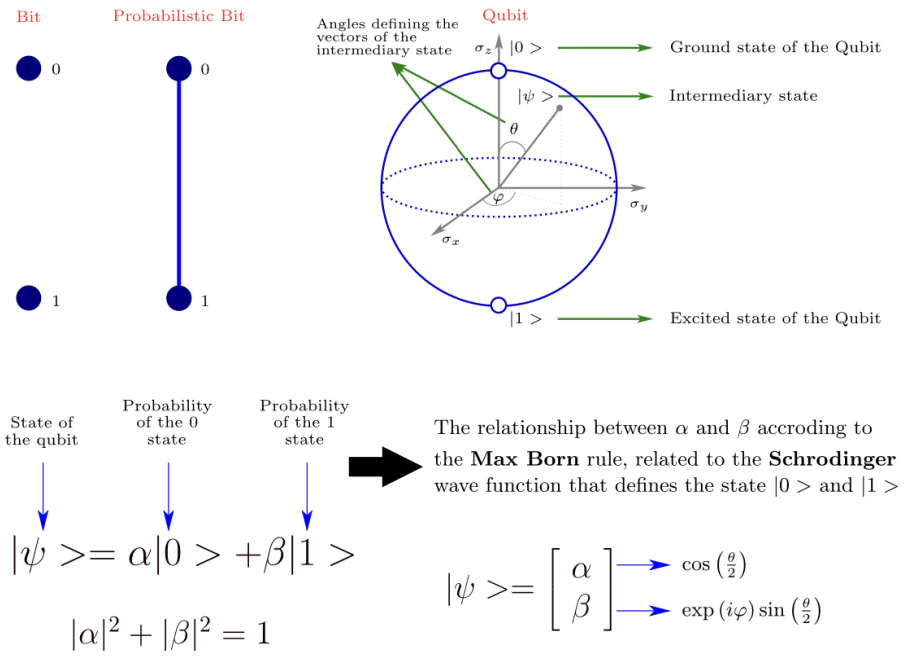


Figure 3.1: Qubits and their mathematical representation taken from Khan et al. [9]

3.2 Quantum Machine Learning

Quantum machine learning (QML) is a field with great potential at the intersection of quantum computing and machine learning, which aims to leverage the unique features of quantum computers to enable new algorithms and approaches for machine learning tasks [11, 12]. The potential of QML to enhance the performance of machine learning algorithms and solve certain problems faster and more efficiently than classical methods has attracted significant attention in recent years [13–15].

One area where QML has shown particular promise is in classification problems, which involve assigning a data point to one of a predefined set of classes based on certain features or attributes. Several studies have explored the use of QML techniques for classification tasks, including the use of quantum neural networks [14], quantum support vector machines [15], and quantum k-nearest neighbors algorithms [16]. These approaches have demonstrated improved performance compared to classical methods, particularly for high-dimensional data sets [9].

However, the development and implementation of QML models remains a challenge due to the limited availability of quantum hardware and the noise and errors that can occur in quantum systems [1, 17]. As a result, researchers are also exploring the use of quantum simulation and noisy intermediate-scale quantum (NISQ) devices for QML [18, 19].

3.3 Motivation

The motivation for studying quantum machine learning lies in the unique features of quantum computers that can potentially revolutionize traditional machine learning techniques. As pointed out by Biamonte et al. [11], quantum computers have the ability to perform certain calculations much faster than classical computers, thanks to the phenomena of superposition and entanglement.

This has led to the development of new algorithms and approaches for machine learning tasks using quantum computers [12]. Moreover, as highlighted by Houssein et al. [13], quantum machine learning has the potential to address some of the limitations of classical machine learning, such as the need for large amounts of labeled data and the difficulty in dealing with high-dimensional data. Quantum machine learning algorithms can also potentially improve the generalization ability of machine learning models, allowing them to make more accurate predictions on unseen data [14]. Furthermore, the application of quantum machine learning has the potential to solve real-world problems in various domains, such as healthcare and optimization problems in finance, logistics and transportation [11, 15]. For example, quantum machine learning algorithms have been used to classify medical images and predict the likelihood of a patient developing a certain disease [19].

However, despite these promising prospects, the field of quantum machine learning is still in its early stages and faces several challenges. As noted by Selig et al. [1], the use of actual quantum devices is limited by their noise and limited number of qubits, which can affect the performance of quantum machine learning algorithms. Additionally, there is a lack of application-oriented benchmarks for evaluating the performance of quantum machine learning algorithms, as discussed by Lubinski et al. [17]

The use of QML for classification problems has also been a focus of research, as discussed in [15]. In [9], the authors compare the performance of quantum and classical models for classification tasks and find that quantum models can potentially achieve higher accuracy and faster training times. However, the authors also note that the development of quantum models for classification tasks is still in its infancy and there is a need for further research to optimize and improve these models. The development and evaluation of QML models also requires the use of either simulation or actual quantum devices, as discussed in [17]. In [1], the authors argue for the use of noisy, shallow gate-based quantum circuits in QML, as they can be more easily implemented on current quantum hardware.

In essence, the motivation for research in the field of QML lies in the potential for quantum computers to solve problems that are currently intractable for classical computers and to advance machine learning. Further research is needed to optimize and improve QML models, as well as to better understand the potential and limitations of using quantum computers for machine learning tasks.

3.4 Quantum Machine Learning vs. Classical Machine Learning

In the study by Houssein et al. [13], the authors compare the performance of quantum machine learning algorithms with their classical counterparts for various tasks such as classification, clustering, and dimensionality reduction. They found that quantum machine learning algorithms generally outperform classical algorithms in terms of accuracy and speed, especially for large-scale, imbalanced, and high-dimensional datasets. For example, they showed that a quantum version of the k-nearest neighbors algorithm achieved a higher accuracy than the classical version for a high-dimensional data set. They also found that quantum algorithms can be used to improve the performance of ensembles of classical models, such as boosting and bagging.

In their study, Lubinski et al. [17] compare the performance of quantum and classical machine learning algorithms for a range of real-world applications. They again found that quantum algorithms can achieve significant improvements in terms of accuracy and speed compared to classical algorithms, especially for large-scale datasets.

Selig et al. [1] also compare the performance of quantum and classical machine learning algorithms and find that quantum algorithms can achieve significant improvements in terms of accuracy and convergence speed.

Mezquita et al. [16] conduct a review of k-nearest neighbor algorithms based on classical and quantum machine learning and find that quantum algorithms can achieve significant improvements in terms of accuracy and convergence speed compared to classical algorithms, especially for large-scale datasets.

Khan and Robles-Kelly [9] compare the performance of quantum and classical machine learning algorithms for various tasks such as classification, clustering, and dimensionality reduction. They find that quantum algorithms can achieve significant improvements in terms of accuracy and convergence speed compared to classical algorithms, especially for large-scale and high-dimensional datasets.

However, Houssein et al. [13], Lubinski et al. [17], Selig et al. [1], Mezquita et al. [16], and Khan and Robles-Kelly [9] all noted that the performance of quantum machine learning algorithms can be sensitive to noise and decoherence, which can impact the accuracy of the results.

Wiebe et al. [14] demonstrated the potential of quantum machine learning for improving the accuracy and efficiency of deep learning algorithms. They used a quantum version of the popular backpropagation algorithm and showed that it can outperform its classical counterpart in terms of convergence speed and accuracy on a number of datasets.

Abohashima et al. [15] conducted a survey of classification techniques using quantum machine learning and found that quantum algorithms can achieve higher accuracy and faster convergence than classical algorithms for certain types of datasets. They also identified some challenges that need to be addressed in order to further improve the performance of quantum machine learning algorithms, such as the limited size of currently available quantum devices and the need for robust error correction techniques.

In their review, Schuld et al. [12] discuss the potential of quantum machine learning for improving the accuracy and efficiency of machine learning algorithms. They provide examples of quantum algorithms that have been proposed for various tasks such as classification, clustering, and dimensionality reduction, and discuss the challenges and limitations of quantum machine learning.

Biamonte et al. [11] also explore the potential of quantum machine learning for improving the performance of machine learning algorithms. They demonstrate that quantum algorithms can be used to achieve a quadratic speed-up over classical algorithms in certain cases. For example, they show that a quantum version of the popular support vector machine algorithm can achieve a speed-up of up to $O(n^2)$, where n is the number of training examples, compared to the classical $O(n^3)$ complexity. They also demonstrate that quantum algorithms can achieve a similar speed-up for supervised and unsupervised learning tasks.

Savchuk and Shapovalova [19] also investigated the use of quantum machine learning for classification problems. They found that quantum algorithms can achieve improved performance over classical algorithms for certain data sets, but the benefit is more pronounced for smaller data sets. They also found that quantum algorithms can be used to improve the interpretability of classical models, such as by providing a more compact representation of the decision boundary.

Petar Jurcevic et al. [4] highlights the potential advantages of using a quantum enhanced support vector machine algorithm. They find that the QSVM was able to perform computations in exponentially large feature spaces over its classical counterpart, which makes it ideal for large-scale data analysis problems. However, they highlight a number of challenges that need to be overcome before scalable and practical quantum computing applications for machine learning are achieved.

In their paper, Schuld and Killoran [2] introduce a novel approach to encode classical data into quantum states that can be utilized to train quantum machine learning models. The proposed method aims to enhance the classification accuracy of a quantum support vector machine by allowing the quantum system to access a larger feature space than is possible in classical machine learning. They demonstrate the potential of their technique by testing it on several benchmark classification tasks. Their findings show that a quantum support vector machine trained with quantum-enhanced feature spaces outperforms a classical support vector machine.

Havlivcek et al. [20] also explore the potential of quantum-enhanced feature spaces for improving the performance of supervised quantum learning algorithms. Havlivcek et al were able to show that quantum-enhanced feature spaces, when combined with classical preprocessing techniques, can be used to improve the performance of classical machine learning algorithms, for instance a classical support vector machine for instance.

Babbush et al. [3] present a quantum computing approach which they call the Variational Quantum Eigensolver (VQE), for solving problems in chemistry and condensed matter physics. The VQE algorithm is a form of variational quantum circuit, and designed to find the ground state of a given Hamiltonian. Their analysis demonstrates that the VQE approach can solve problems that are hard for classical computing methods to deal with.

In conclusion, these studies suggest that quantum machine learning can offer improved performance over classical algorithms in certain cases, particularly for high-dimensional and imbalanced data sets. However, the benefit of quantum algorithms is not always clear and can be sensitive to factors such as noise and the specific data set and task. As such, it is important to carefully evaluate the performance of quantum machine learning algorithms and consider the trade-offs between their potential benefits and the challenges of implementing and scaling them.

3.5 Current approaches to Quantum Machine Learning

Quantum machine learning has gained significant attention in recent years due to its potential to improve upon traditional machine learning techniques. There are various approaches to quantum machine learning, including quantum-inspired classical algorithms, hybrid classical-quantum algorithms, and fully quantum algorithms.

Quantum-inspired classical algorithms aim to replicate the behavior of quantum systems using classical computational methods. Examples of these include quantum support vector machines and quantum neural networks. Quantum support vector machines have been applied to classification tasks and demonstrated improved performance over their classical counterparts in some cases [9].

Quantum neural networks, inspired by classical neural networks, have been used for tasks such as image recognition and quantum state classification [13, 15]. These algorithms make use of quantum mechanical phenomena, such as superposition and entanglement, which may lead to more efficient and accurate machine learning algorithms [11, 12].

Hybrid classical-quantum algorithms utilize both classical and quantum computational resources. One example is the quantum k-nearest neighbors (QKNN) algorithm, which combines classical k-nearest neighbors classification with quantum computing. The QKNN algorithm has been found to outperform classical KNN in terms of accuracy and runtime, particularly in high-dimensional feature spaces [16]. Quantum algorithms have also been explored for feature selection, the process

of selecting relevant features from a larger set to improve machine learning model performance. Quantum algorithms have been applied to various data sets and have demonstrated effectiveness in this task, including for biological data [18].

A fully quantum machine learning algorithm is a type of algorithm that is designed to be run entirely on a quantum computer, using quantum gates and quantum circuits to perform calculations that are not possible on classical computers [11]. One example of a fully quantum algorithm is quantum principal component analysis (QPCA), which is a quantum version of the classical principal component analysis (PCA) algorithm. QPCA has been applied to a variety of data sets and has been shown to have improved performance over classical PCA in terms of both accuracy and runtime [11]. However, the use of QPCA is limited by the current size and noise levels of quantum computers, which may prevent it from being practical for certain tasks [17].

Quantum variational classifiers (VQCs) are a class of fully quantum algorithms that can be used for classification tasks. These algorithms make use of superposition and entanglement to achieve a more efficient classification process than classical machine learning algorithms. VQCs have been applied to a variety of datasets and have been shown to be effective in classification tasks [20]. However, the performance of VQCs is also limited by the size and noise levels of current quantum computers, and further research is needed to improve their performance and scalability.

Another example of a fully quantum algorithm is quantum support vector machines (QSVMs), which are based on the classical support vector machine algorithm but are implemented using quantum mechanics. QSVMs have been applied to a variety of data sets and have been shown to be effective for classification tasks [9]. However, like other fully quantum algorithms, the performance of QSVMs is limited by the current size and noise levels of quantum computers, as well as the need for error correction techniques [1].

In conclusion, quantum machine learning is a rapidly developing field that has the potential to revolutionize traditional machine learning techniques. There are a wide range of approaches to quantum machine learning, including quantum-inspired classical algorithms, hybrid classical-quantum algorithms, and fully quantum algorithms. While quantum machine learning has shown promise for a variety of tasks, it also faces significant challenges due to the fragile nature of quantum systems and the need for error correction techniques. As a result, further research is needed to fully understand the potential benefits and limitations of fully quantum algorithms for machine learning tasks.

3.6 Simulated Quantum Machine Learning performance compared to a real device

Simulated quantum machine learning algorithms refer to those that are run on classical computers, using software libraries or simulators to replicate the behavior of quantum systems. These algorithms can provide a useful tool for exploring the potential benefits and limitations of quantum machine learning without the need for access to a physical quantum computer.

However, the performance of simulated quantum machine learning algorithms may not always accurately reflect the performance of the same algorithms on a physical quantum device. This is because simulations may not perfectly replicate the behavior of real quantum systems, which are subject to noise and other imperfections [17]. As a result, it is important to consider the limitations of simulated algorithms when evaluating the potential of quantum machine learning.

One study that compared the performance of simulated and actual quantum algorithms for machine learning tasks was conducted by Selig et al. [1]. The authors found that, in some cases, simulated algorithms performed similarly to their counterparts on an actual quantum device, but in other cases there were significant differences in performance. The authors concluded that the use of actual quantum devices is important for accurately evaluating the potential of quantum machine learning algorithms.

Other studies have also found that the performance of quantum machine learning algorithms can vary significantly depending on the specific task and data set being used [13]. As a result, it is important to carefully evaluate the performance of quantum machine learning algorithms on a variety of tasks and data sets, both in simulation and on actual quantum devices, in order to fully understand their potential and limitations.

In summary, simulated quantum machine learning algorithms can provide a useful tool for exploring the potential of quantum machine learning, but it is important to consider the limitations of these simulations when evaluating the performance of quantum algorithms. Actual quantum devices are necessary for accurately evaluating the potential of quantum machine learning algorithms, and the performance of these algorithms may vary depending on the specific task and data set being used.

3.7 Specific Methods

The Quantum k-nearest neighbors (QKNN) algorithm is a hybrid classical-quantum machine learning approach that combines the classical k-nearest neighbors (KNN) classification algorithm with quantum computing [16]. The QKNN algorithm is designed to classify data points based on their proximity to other data points in a feature space. In the QKNN algorithm, the distance between data points is calculated using the quantum version of the Euclidean distance, which is defined as the square root of the sum of the squares of the differences between the corresponding elements of two vectors [16].

The QKNN algorithm operates in two main stages: a quantum stage and a classical stage. In the quantum stage, the algorithm uses a quantum circuit to perform a quantum version of the KNN classification, which is based on the quantum version of the Euclidean distance. In the classical stage, the algorithm uses a classical computer to perform the final classification based on the results of the quantum stage.

One of the key advantages of the QKNN algorithm is its ability to handle high-dimensional feature spaces, which are often difficult to process using classical KNN algorithms due to the "curse of dimensionality" [16]. QNN algorithms have been shown to have improved performance over their classical counterparts in this scenario [21]

Another key advantage of the QKNN algorithm is that it can make use of quantum mechanics to find the nearest neighbours more efficiently than classical KNN algorithms. This is achieved through the use of quantum search algorithms, such as Grover's algorithm, which can find a specific item in an unsorted list of items in a quadratic speedup compared to classical search algorithms [16].

The QKNN algorithm also utilises quantum feature maps, which encode the feature vectors into quantum states. This allows for the distance between the feature vectors to be calculated using quantum distance measures, such as the quantum Jaccard distance or the quantum overlap distance [16].

In terms of implementation, the QKNN algorithm can be run on both simulated quantum computers

and actual quantum devices. Studies have shown that the performance of the QKNN algorithm is improved on actual quantum devices, especially in high-dimensional feature spaces [16], although the use of actual quantum devices is limited by the current limitations of quantum hardware, such as the need for error correction and the limited number of qubits available [17]. In addition, the QKNN algorithm has been shown to have improved performance in terms of both accuracy and runtime compared to classical KNN algorithms.[16].

Another method for implementing a Quantum Machine Learning algorithm is the use of quantum convolutional neural networks (QCNNs), which are a type of QNN that is based on classical convolutional neural networks (CNNs), which are a type of neural network that is specifically designed for image recognition and other tasks involving data with a spatial structure. [15]. However, QCNNs are implemented using quantum mechanical principles, such as superposition and entanglement, which allows them to take advantage of the unique features of quantum systems and potentially lead to improved performance [13].

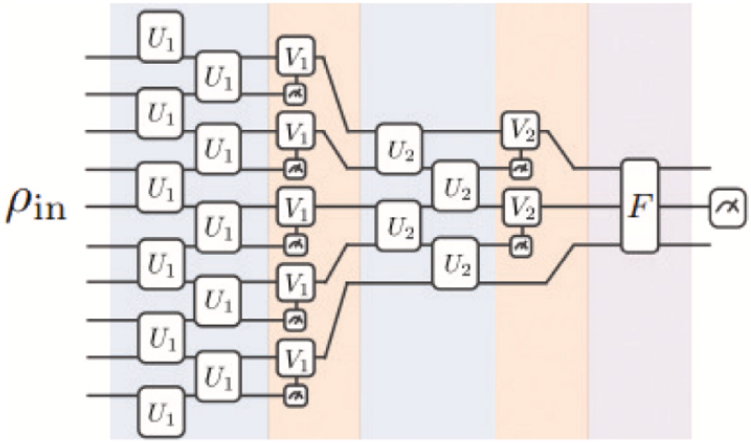


Figure 3.2: The quantum convolutional neural network proposed by Cong et al. [22]

There is also quantum support vector machines (QSVM) [4] as explained in 3.5. QSVMs are based on the classical support vector machine (SVM) algorithm, which is widely used for classification tasks. QSVMs make use of quantum feature maps to encode classical data into quantum states and apply quantum algorithms to perform the classification task. Similarly, quantum variational classifiers (VQCs) have also been proposed, which are a type of quantum circuit that can be trained to perform classification tasks [2]. VQCs can be trained using classical optimization algorithms, and their performance has been shown to improve with increasing quantum resources. Both QSVMs and VQCs have the potential to outperform classical SVMs in terms of both accuracy and efficiency, especially in high-dimensional feature spaces, although more studies are needed to fully assess their capabilities [4].

In addition to these methods, there have been a number of studies exploring the use of quantum computers for training classical machine learning models, such as deep neural networks, using quantum algorithms [14]. These studies have shown that quantum computers can potentially provide a speedup for certain tasks, such as training deep neural networks [14]. However, the performance of these approaches is dependent on the specific task and data set, and more research is needed to fully understand the potential benefits and limitations of using quantum computers for training classical machine learning models [13].

To summarise, there are a number of different approaches to implementing Quantum KNN and Quantum Neural Networks, each with their own advantages and challenges. Further research is needed to fully understand the potential benefits and limitations of these methods, as well as to develop new approaches that can take advantage of the unique features of quantum systems.

3.8 Challenges

There are a number of challenges associated with implementing and using quantum machine learning algorithms. One of the main challenges is the limited availability of quantum hardware, which makes it difficult to test and evaluate quantum machine learning algorithms in practice. Additionally, quantum systems are fragile and susceptible to noise, which can negatively impact the performance of quantum machine learning algorithms [13]. As part of our objectives we utilised the Azure Quantum Cloud Computing platform [23].

Another challenge is the lack of a standardized framework for implementing and evaluating quantum machine learning algorithms. This makes it difficult to compare the performance of different algorithms and to determine the optimal approaches for specific tasks [17]. By using a number of evaluation measures, as well as comparing performance across multiple datasets, we can evaluate the performance of our algorithms on a number of metrics in order to meet the project's evaluation objectives

In addition to the problem of the hardware availability when implementing quantum machine learning algorithms, there is limited resources available on the current quantum devices themselves. Quantum computers have a limited number of qubits, which limits the size and complexity of the algorithms that can be run on them. In addition, quantum systems are prone to errors and noise, which can degrade the performance of quantum algorithms and require the use of error correction techniques [11, 17].

One approach to addressing these challenges, and the one we will be taking while developing our algorithms, is the use of shallow circuits, which are quantum circuits that are designed to be simple and require fewer resources. Shallow circuits have been shown to be effective for a number of quantum machine learning algorithms, including quantum neural networks and quantum support vector machines [1]. The use of shallow circuits can potentially improve the performance and scalability of quantum machine learning algorithms, as well as make them more feasible to run on current quantum devices [1]. However, it is important to note that the trade-off for using shallow circuits is that they may have lower accuracy compared to deeper circuits [1].

In addition, there are challenges associated with the development of quantum machine learning algorithms themselves. For example, there is a need for new quantum algorithms that are specifically designed for machine learning tasks, as well as for classical machine learning algorithms that can take advantage of the unique features of quantum systems [13].

Ultimately, there are a number of challenges associated with the implementation and use of quantum machine learning algorithms, which require further research and development in order to fully realize the potential of quantum machine learning.

Chapter 4: Data Considerations

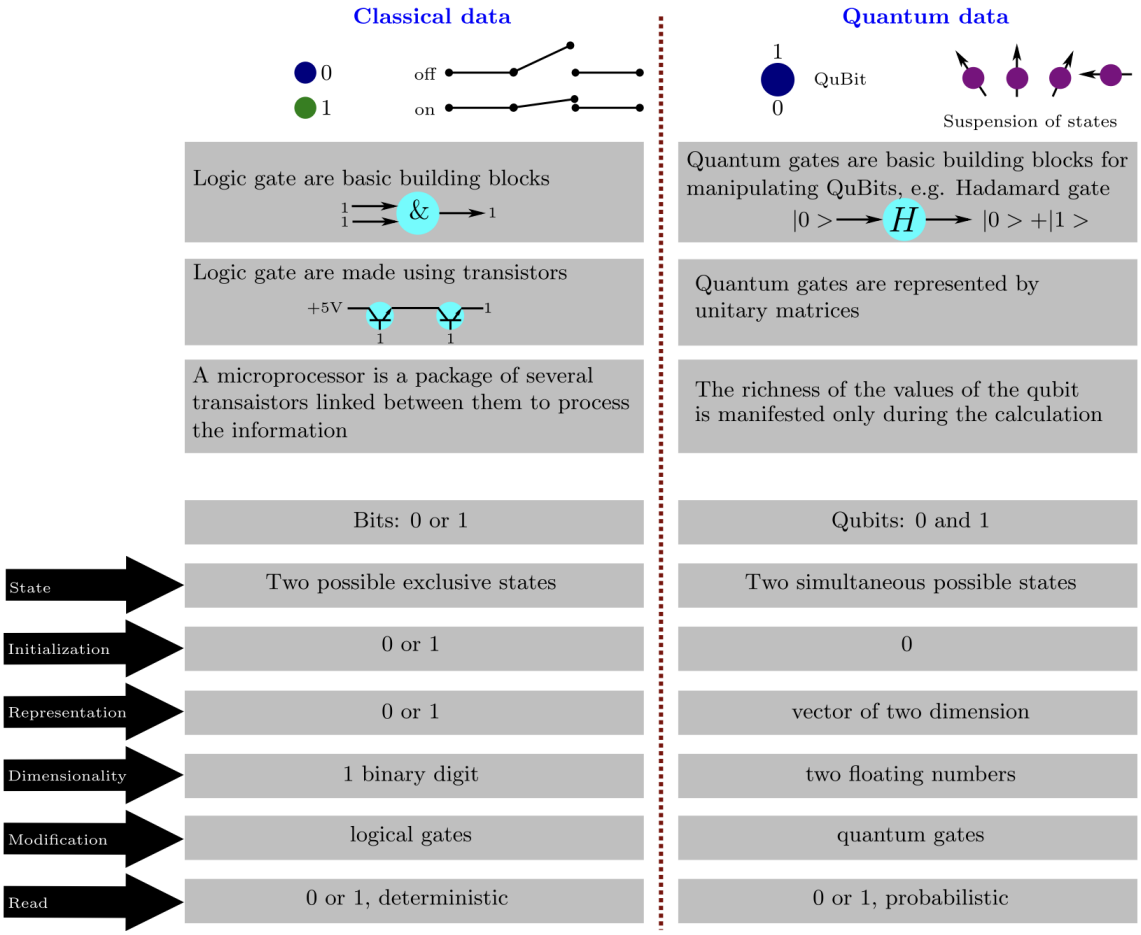


Figure 4.1: Comparison between classical and quantum data taken from Khan et al. [9]

When planning our project, we need to consider the nature and scale of the data that will be used to test the quantum machine learning algorithms, especially as the dataset used can greatly affect the performance of a quantum machine learning algorithm as explained in Sections 3.4 and 3.8 of Chapter 3. One important factor to consider is the cardinality of the classes in the dataset. Quantum machine learning algorithms may be more effective for datasets with a smaller number of classes, as they can become more complex and computationally expensive when dealing with a larger number of classes [11]. This complexity can be further exacerbated when breaking up the bit string output as well. Another factor to consider is the number of features in the dataset. Quantum machine learning algorithms are typically limited by the number of qubits available, and as such, datasets with a large number of features may not be well-suited for testing [17] so in response, the design of our algorithms will assign one qubit per feature to reduce complexity. It is generally recommended to use datasets with 8 or fewer features when testing quantum machine learning algorithms [16].

In terms of data types, it is important to consider whether the dataset contains mixed data types, such as both continuous and categorical variables. Quantum machine learning algorithms may

be more challenging to implement for datasets with mixed data types [18], and it may be more effective to use datasets with a single data type [13]. To ensure the suitability of the data for our project, we will carefully evaluate the characteristics of the dataset and select a dataset that is well-suited for testing the performance of the quantum machine learning algorithms.

Once the appropriate dataset has been selected, we will need to convert the data into a format that the quantum machine learning algorithms can use. One method is to use classical preprocessing techniques to convert the data into a numerical format, such as integers or floating point numbers, and scale the data to a specific range [14]. The processed data can then be input into the quantum computer using qubits, which are the basic units of quantum information [11]. Alternatively, we can use a quantum algorithm to directly process the raw data and convert it into a form that can be used by the quantum computer [18]. This approach may be more suitable for certain types of data, such as categorical or discrete data, which may be difficult to represent using classical numerical formats [17].

Once we have carefully considered the data considerations for our project we can narrow down our possible candidates to a few commonly used UCI datasets commonly used for classification such as the Iris, Wine Quality, and Ecoli datasets [24]. Then by converting the data into a suitable format, we can ensure that the quantum machine learning algorithms are able to accurately and effectively process the data.

Chapter 5: Outline of Approach

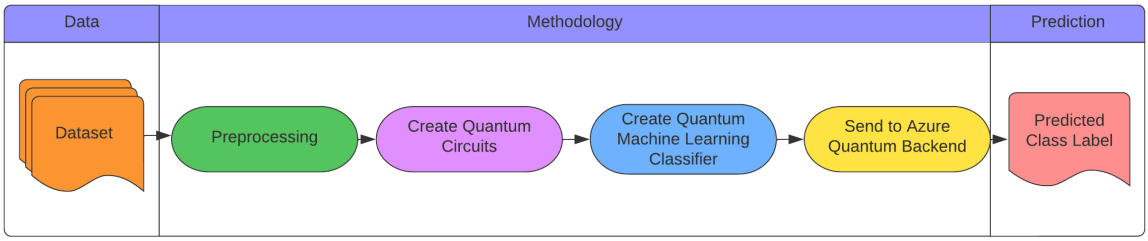


Figure 5.1: Overview of Methodology for the prediction of class labels

In this project, we aim to compare the performance of a quantum support vector machine (QSVC) and a variational quantum classifier (VQC) for various classification tasks as against their classical equivalents, these quantum machine learning models were chosen due to their ability to utilise the unique properties of quantum computing, such as superposition and entanglement, and offer enhanced computational capabilities and the ability to explore exponentially large feature spaces [2, 20].

The QSVC algorithm, described in [4], is a hybrid classical-quantum approach that combines classical support vector machine (SVM) classification with quantum computing. It operates by encoding data points into quantum states and using quantum gates and circuits to build upon a classical SVM kernel and create a quantum version of the SVM kernel function. The resulting kernel matrix is used to train a classical SVM and classify new data points. The QSVC algorithm has the advantage of being able to handle high-dimensional feature spaces, which can be challenging for classical SVM algorithms due to the curse of dimensionality [20].

For the VQC, we will to use a quantum circuit-based approach, as described by [2], by utilizing either the Tensorflow Quantum [25] or Pennylane's in-built libraries [26] as a base implementation. VQCs are a type of QNN that use quantum circuits to perform learning and inference. They are based on the concept of variational quantum algorithms, which use variational principles to find the optimal parameters for a quantum circuit that minimises the cost function of a given classification task.

To test the performance of these algorithms, we will use various datasets with different characteristics, including the number of features and data types. We will evaluate the performance of the QSVC and VQC algorithms in terms of accuracy and runtime, and compare the results to those obtained using classical machine learning algorithms.

In order to optimize the performance of our algorithms, we will be carefully selecting the dataset that we use for testing in order to meet challenges identified in our discussion of the data consideration in Chapter 4, such as the number of features in the dataset, and availability of qubits.

Once the data is prepared, we will use the Azure Quantum Cloud Computing platform [23] to test our algorithms. Azure Quantum Cloud Computing is a cloud-based platform that allows users to access a number of real quantum computers and run quantum algorithms. We will use the platform to send our code and data to the quantum computers, and receive the results of the computations. This will allow us to compare the performance of our quantum machine learning algorithms to that of classical algorithms and evaluate the potential advantages of using quantum computing for machine learning tasks.

In summary, our approach to this project involves implementing and comparing the performance of a quantum support vector machine (QSVC) algorithm and a variational quantum classifier (VQC) using various datasets and the Azure Quantum Cloud Computing platform. By carefully considering the characteristics of the datasets and optimising the implementation of our algorithms, we hope to demonstrate the potential advantages of using quantum computing for machine learning tasks.

Chapter 6: Project Workplan/Implementation

6.1 Project Workplan

The project work plan, as shown in Figure 6.1 below, consists of six sprints, with five sprints and a buffer and backlog. The duration of each sprint is measured in weeks to allow for flexibility in light of college assignments and other time-consuming elements of the project, such as testing algorithms on Azure Quantum. By organizing the project into sprints with Sprint 4 overlapping the other sprints, we can effectively manage the various uncertainties that may arise and ensure that the project stays on track. Overlapping the sprints allows us to continue making progress on the project while waiting for results from IBM or other quantum platform we use, which helps to avoid delays caused by waiting for significant parts of the project to be completed. The output of each sprint are also indicated in order to track progress throughout the project.

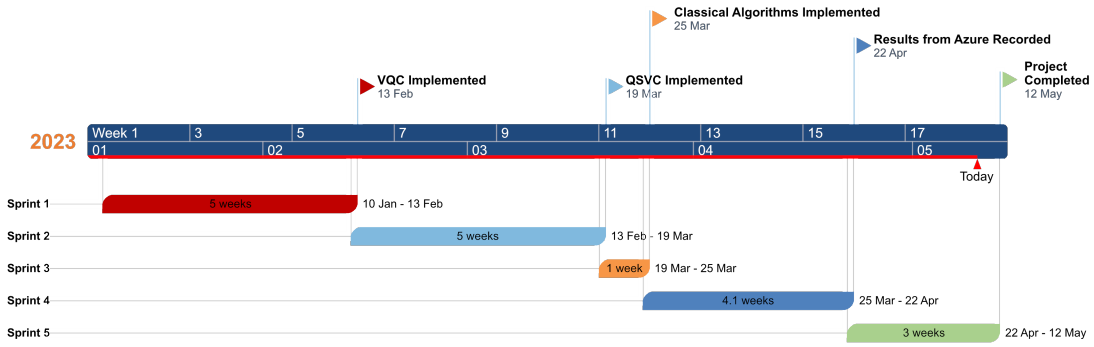


Figure 6.1: Gantt Chart of Project Work Plan

Sprint 1 - The first sprint of this project will focus on downloading required libraries, establishing a development environment, creating a simple dataset for testing, and creating a simulated VQC algorithm. This sprint will take approximately five weeks to complete. The VQC algorithm is a relatively simple Quantum Machine Learning Classification Algorithm that we will use for testing purposes.

Sprint 2 - The second sprint of this project will involve developing a simulated QSVC algorithm. This sprint will take approximately five weeks to complete. The QSVC algorithm is another relatively simple Quantum Machine Learning Classification Algorithm that we will also use for testing purposes.

Sprint 3 - The third sprint of this project will involve comparing the simulated Quantum algorithms from the previous sprints with their classical equivalent algorithms. We will observe and compare

the performance of all the algorithms. This sprint will take approximately one week to complete.

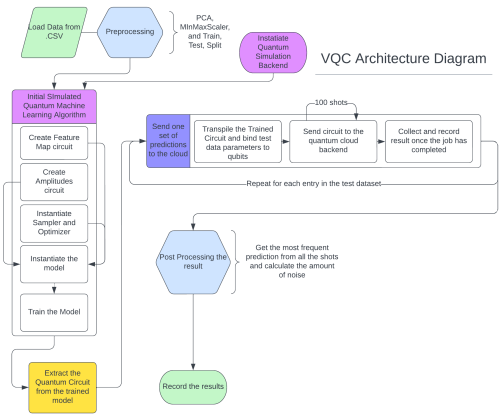
Sprint 4 - The fourth sprint of this project will involve testing the QSVC and Quantum VQC algorithms on a real device, using Azure Quantum (or another platform). This is the most time-consuming aspect of the project, as it involves sending the code to IBM Quantum and waiting for the results, which may take several weeks. As a result, we have allocated ten weeks for this sprint. While waiting for the results, we will work on other sprints. Once we have the results back from Azure Quantum we can observe and compare the performance with the simulated algorithms and the classical algorithms.

Sprint 5 - The fifth and final sprint of this project will focus on evaluating all of the results from the previous sprints and writing the final report. This will include creating any necessary graphics for the report and presentation. This sprint will take approximately three weeks to complete, as all of the technical aspects of the project will have been completed at this point.

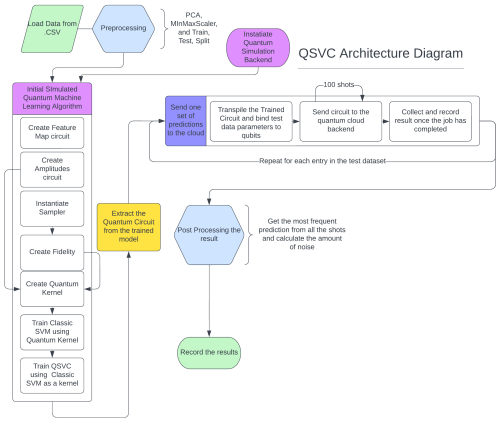
Buffer and Backlog - Buffer and Backlog is not shown on the Gantt chart because it will take place during any available time between the completion of Sprint 5 and the submission date, and will be used to make any additional improvements or additions to the final report. This will allow us to ensure that the final report is as comprehensive and polished as possible.

6.2 Implementation of QSVC and VQC

The implementation of Quantum Support Vector Classification (QSVC) and Variational Quantum Classifier (VQC) involves combining quantum computing techniques with classical machine learning algorithms. These hybrid models leverage the power of quantum computation for specific tasks while incorporating classical pre-processing and post-processing steps.



(a) Architecture Diagram of the VQC Implementation



(b) Architecture Diagram of the QSVC Implementation

Figure 6.2: Architecture Diagrams the different quantum implementations

6.2.1 QSVC Implementation

QSVC is based on the quantum support vector machine (QSVM) algorithm, which utilizes the quantum kernel method to classify data points. The implementation of QSVC involves several key steps:

Quantum Feature Mapping: The classical input data is encoded into a quantum state using a feature mapping circuit. This circuit maps classical data points to quantum states by employing techniques such as amplitude encoding or quantum embeddings. We use amplitude encoding due to its ability to make use of the inherent parallelism in quantum computing, and its compatibility with variational algorithms such as QVC and QSVC.

Quantum Kernel Evaluation: The quantum feature mapping is followed by the evaluation of the quantum kernel. This step involves comparing the similarity between pairs of quantum states using quantum operations. Quantum kernel methods, such as the quantum inner product, are utilized to compute the similarity matrix.

Quantum Optimization: The quantum kernel matrix is then used as input for a classical optimization algorithm, such as the Sequential Minimal Optimization (SMO), to find the support vectors and optimize the classification hyperplane. This step involves iteratively updating the weights and biases of the classifier to minimize the classification error.

Quantum Measurement and Prediction: Once the classification hyperplane is optimized, the quantum state representing the test data is prepared using the same feature mapping circuit. The resulting quantum state is measured, and the measurement outcome is used to predict the class label of the test data point.

Extracting the trained Quantum Feature Mapping: After training and testing the feature mapping circuit, it is extracted, transpiled, and deployed on a real Quantum device via the Azure Quantum Computing Cloud Platform. This step allows for the practical implementation and execution of the trained circuit.

Quantum Prediction and Post-Processing: In the prediction phase, the labels are bound to the quantum gates and the circuit is executed for a specified number of shots, typically 100, we do this in order to work around noise. 100 shots were chosen for our experiments as it would allow for easy percentage calculations while reducing computation time. After discarding returns that do not conform to the expected format (e.g., noise), the most frequently observed return is considered as the predicted class label for the corresponding test data point. This post-processing step ensures reliable and accurate classification based on the outcomes of the quantum circuit execution.

6.2.2 QVC Implementation

The Variational Quantum Classifier (VQC) is a hybrid machine learning model that combines classical optimization techniques with variational quantum circuits. The implementation of VQC involves several key steps:

Quantum Circuit Design: A variational quantum circuit is designed to represent the trainable parameters of the model. This circuit consists of a series of quantum gates, including single-qubit rotations and two-qubit entangling gates. The specific structure and depth of the circuit may vary based on the problem and available resources.

Quantum State Preparation: The variational quantum circuit is used to prepare a quantum state that encodes the input data. The data is usually mapped to the amplitudes or other properties of the quantum state using encoding techniques suitable for the specific problem as explained above in 6.2.1

Classical Optimization: A classical optimization algorithm, such as gradient descent or the Nelder-Mead method, is employed to update the parameters of the variational quantum circuit, we chose the COBYLA (Constrained Optimization by Linear Approximation) algorithm as the

optimizer for the VQC in our study due to its suitability for handling constrained optimization problems. The objective is to minimize the cost function, which measures the discrepancy between the predicted and target outputs. COBYLA is particularly suited to problems where the cost function evaluations are noisy as it does not rely on gradient information, this makes it ideal for quantum operations which are run on noisy quantum devices. COBYLA also allows us to put bounds on circuit operations and restrictions on gate operations.

Iterative Training: The optimization process is repeated iteratively, with the quantum circuit parameters updated in each iteration based on the gradients computed from the cost function. The number of iterations and convergence criteria may vary depending on the problem complexity and convergence behavior.

Prediction and Inference: Once the optimization process converges, the trained variational quantum circuit can be used to make predictions on new, unseen data points. The input data is encoded into the quantum state using the same encoding technique used during training, and the resulting quantum state is passed through the trained circuit to obtain the predicted output.

Extracting the trained Quantum Feature Mapping: After training and testing the feature mapping circuit, it is extracted, transpiled, and deployed on a real Quantum device via the Azure Quantum Computing Cloud Platform. This step allows for the practical implementation and execution of the trained circuit.

Quantum Prediction and Post-Processing: In the prediction phase, the labels are bound to the quantum gates and the circuit is executed for a specified number of shots, similar to the QSVC. After discarding returns that do not conform to the expected format (e.g., noise), the most frequently observed return is considered as the predicted class label for the corresponding test data point. This post-processing step ensures reliable and accurate classification based on the outcomes of the quantum circuit execution.

Chapter 7: **Results**

7.1 Evaluation

As Quantum Machine Learning is still a developing field of research, there is still ongoing research and debate over how to evaluate Quantum Machine Learning algorithms correctly as pointed out in [27–29]. This is due, in part, to quantum computing's hardware limitations which make it difficult to evaluate.

As we have explained over the course of this paper, Quantum Machine Learning is an evolving field of research, and thus, the correct way to evaluate Quantum Machine Learning algorithms is still an active area of investigation. This is largely due to the limited availability and limitations of current quantum hardware, which makes testing algorithms at scale rather difficult [1]. In addition, quantum hardware suffers from significant noise and error rates further complicating the issue, and making it difficult to distinguish the performance of quantum algorithms from classical algorithms.

Due to these reasons, there are currently no standard quantum evaluation metrics that can be used to compare and evaluate quantum algorithms other than current machine learning algorithms which may or may not be an accurate representation. To address this, there is continuous development of new techniques and metrics to evaluate Quantum Machine Learning algorithms appropriately which makes choosing an evaluation metric challenging. [27, 29]

While the increasing accessibility of quantum computing platforms has opened up the possibility of broader and sophisticated evaluations, there is hope that as the field of Quantum Machine Learning matures more standardised evaluation metrics will be found. Meanwhile, using simple and easy to interpret metrics such as Accuracy and the measurement of Computing Resources as proposed by Huang et al. [30] can help to evaluate the models and measure them against their classical equivalents. We have used these metrics as the main basis of our comparisons.

Limitations in available quantum hardware meant we could only use quantum circuits of 4 qubits, or less. However, by restricting the experiments to four features or qubits, it provided an opportunity to focus on the fundamental aspects of the models and their performance characteristics as well as optimize the utilization of the available resources. The limited qubits also would not restrict the scalability or potential applicability meaning our findings obtained from these experiments can serve as a foundation for further study and can be extended to higher-dimensional datasets or larger-scale quantum systems in future research.

To corroborate our results, we conducted extensive testing across multiple datasets as well as the 3 distinct real quantum devices (IonQ, Quantinuum, and Rigetti) provided by the Azure Quantum Cloud Computing platform. In addition, we created two simulated quantum environments, one with noise and one without, in order to measure the performance of our algorithms under idealised quantum conditions. This allows us to measure our quantum machine learning models ability to handle machine learning tasks in both realistic and controlled scenarios and paint a more comprehensive picture of their capabilities, limitations, and what the future of quantum machine learning might look like.

To ensure our results remained reliable and make our experiments robust against the noisy quantum environments, each model is executed using a fixed number of shots, specifically 100 shots for the experiments conducted in this study. By using multiple shots we can mitigate the impact of noise

on individual measurements and then use the most commonly returned value as our predicted value.

Several considerations led us to choose 100 shots for our experiments. Firstly, it provides a sufficient number of samples for a reliable estimate of the predicted value while facilitating easy percentage calculations, allowing for more easily interpreted results.

Secondly, the choice of 100 shots allows for a balance between accuracy and computational efficiency. An increased number of shots may improve accuracy, it would also significantly increase the computational burden and execution time. By testing each model with 100 shots, we were able to achieve a reasonable trade-off between accuracy and computational feasibility while also providing each model with a consistent standard to be measured against.

The simulated QSVC and VQC algorithms showed promising results in our tests, achieving similar classification accuracy compared to classical SVM and NN algorithms in some cases. However, when testing the algorithms on a real quantum device using Azure Quantum, we found that the performance of the quantum algorithms was significantly worse than the classical algorithms. The main reason for this is the current limitations of quantum hardware, including noise and limited qubit connectivity, which can cause errors and reduce the accuracy of the quantum algorithms.

We compare the performance of four machine learning algorithms, namely neural network, SVM, VQC, and QSVC, on the three datasets: Iris, Wine, and Breast Cancer. For each algorithm, we measured the accuracy, CPU usage, and memory usage.

7.2 Accuracy

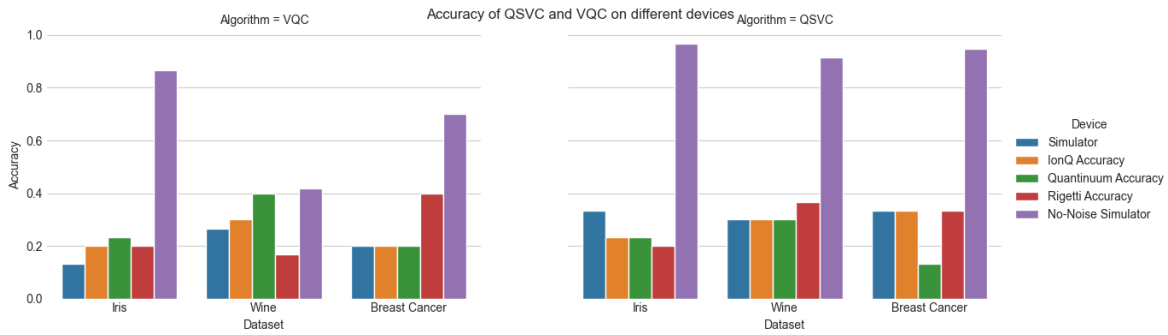


Figure 7.1: Accuracy of the VQC and QSVC on different devices

Overall, the neural network achieved the highest accuracy across all three datasets, with a perfect score of 100% on Iris, 86.1% on Wine, and 90.4% on Breast Cancer. SVM had the second-highest accuracy, with scores of 96.7% on Iris, 86.1% on Wine, and 92.1% on Breast Cancer. The simulated VQC and QSVC reached similar accuracy scored as their classical equivalents with QVC scoring between 86% and 63% and QSVC scoring between 96% and 72%. The VQC and QSVC tested on real quantum devices had lower accuracy scores across all three datasets, with VQC scoring between 13.3% and 40% and QSVC scoring between 30% and 33.3%. However, as we can see from the error bar in 7.2, the consistency of results was varied across the different devices. By this metric Rigetti and Quantinuum were the most consistent devices, followed by IonQ as the most inconsistent real quantum device. The various simulations proved to be quite inconsistent, which is surprising.

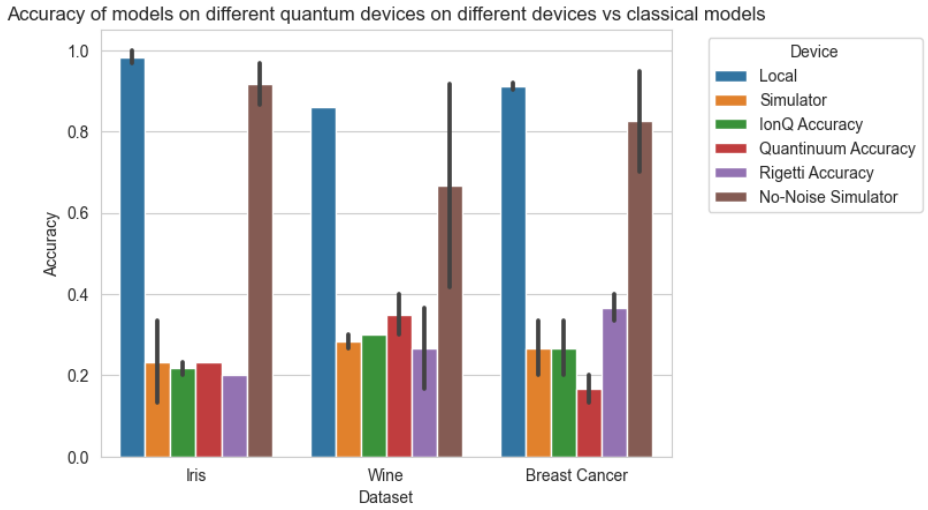


Figure 7.2: Quantum Model accuracy compared to their classical equivalent

7.3 CPU Usage

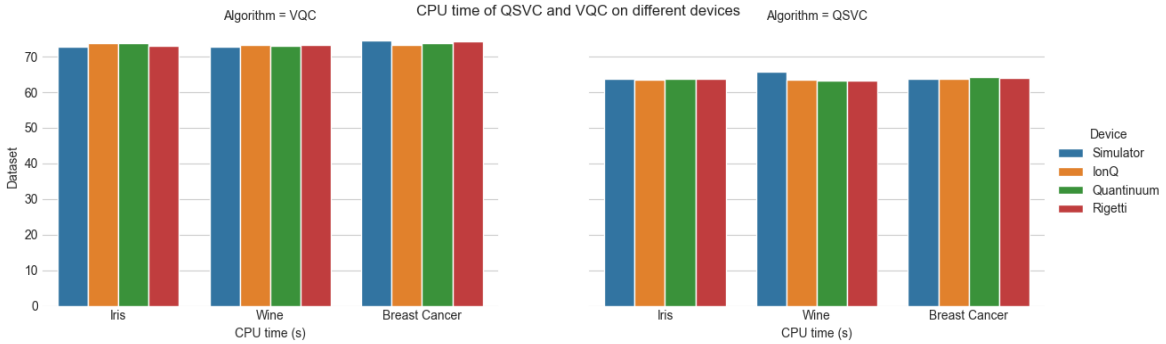


Figure 7.3: CPU Usage of the VQC and QSVC on different devices

For CPU usage, QSVC had the highest usage on all three datasets, with an average of 14.1% for Iris, 14.2% for Wine, and 15.3% for Breast Cancer. Neural Network and SVM had relatively low CPU usage across all three datasets, with neural network averaging between 13.2% and 100%, and SVM averaging between 0% and 50%. VQC had the lowest CPU usage across all three datasets, averaging between 12.6% and 15.3%. As seen in 7.4, the various real Quantum devices were quite consistent in their CPU usage which is unsurprising as they are all run by the Azure Quantum Computing Cloud platform. The locally run algorithms performed quite differently and wildly inconsistent, especially in the case of the Wine Dataset.

7.4 Memory

For memory usage, all algorithms had relatively low memory usage on all three datasets, with an average of around 77-79 MB. Among the quantum algorithms, VQC had slightly higher memory usage compared to QSVC, with an average of around 73-74 MB compared to QSVC's average of around 63-65 MB.

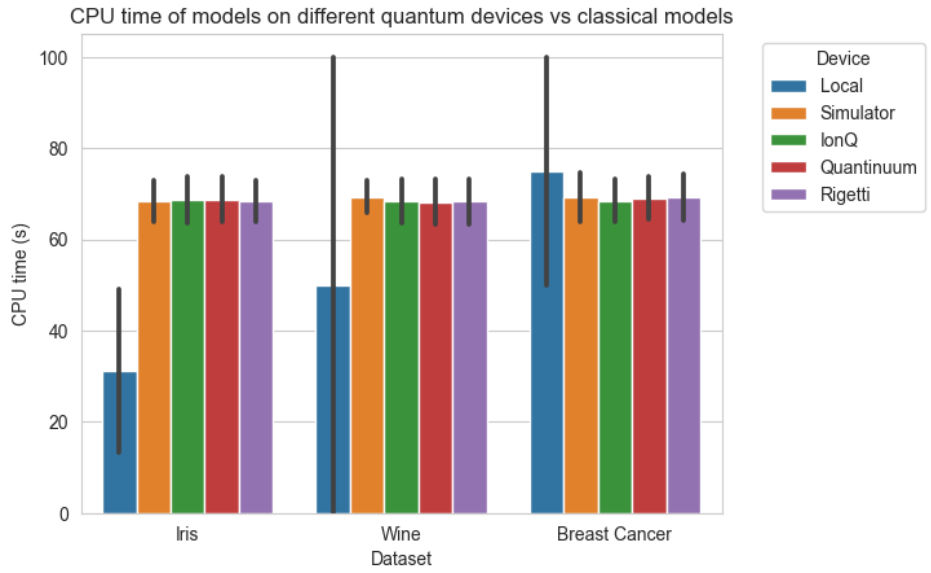


Figure 7.4: CPU time of models on different quantum devices vs classical models

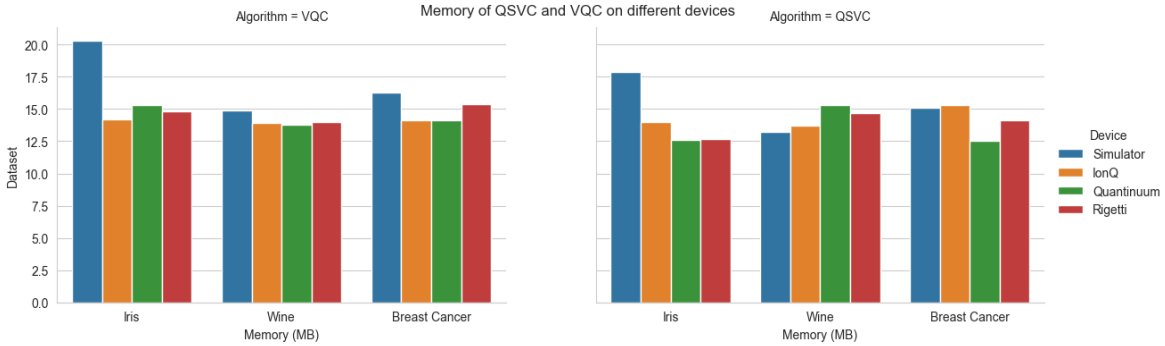


Figure 7.5: Memory of the VQC and QSVC on different devices

7.5 Noise

In our study, the definition of noise was based on returns that did not conform to the expected binary format, which posed challenges in distinguishing between preprocessing and postprocessing steps used in hybrid quantum machine learning models and the trained quantum feature mapping circuit. While this definition of noise may have limitations and is arguably problematic, it provided a framework to evaluate the noise levels in our experiments.

With this definition in mind, we observed that the simulator used for QSVC and VQC had relatively low levels of noise, as evidenced by their high accuracy scores. On the other hand, the actual quantum devices (IonQ, Quantinuum, and Rigetti) showed higher levels of noise, which led to lower accuracy scores. While different devices varied on their noise level, they all managed to have a similar noise of 75%. Showing that the choice of currently accessible device chosen does not have a significant effect on noise.

It is worth noting that noise levels can vary widely depending on the specific quantum device and the type of quantum algorithm being used. In some cases, noise can be mitigated through error-correction techniques, but these techniques can also increase the computational overhead and complexity of the algorithm.

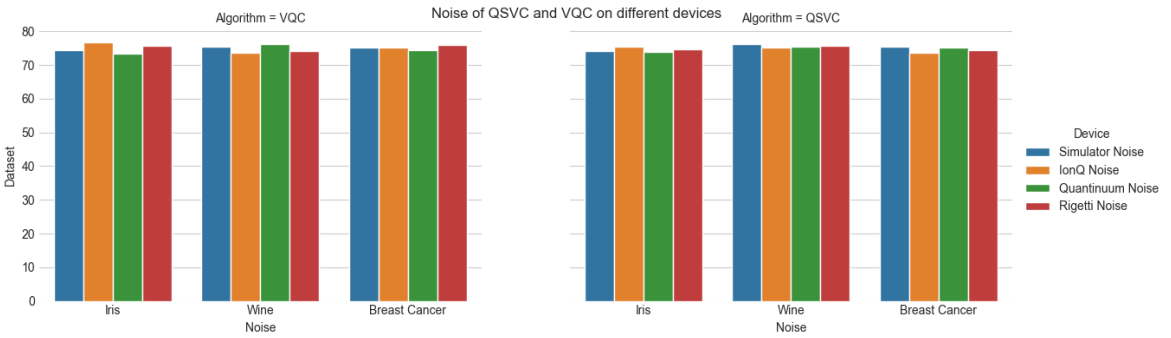


Figure 7.6: Noise of the VQC and QSVC on different devices

7.6 Discussion

The simulated QSVC and VQC algorithms showed promising results, achieving similar classification accuracy compared to classical SVM and Neural Network algorithms in some cases. However, when testing the algorithms on real quantum devices using Azure Quantum, we found that the performance of the quantum algorithms was significantly worse than the classical algorithms. The main reason for this is the current limitations of quantum hardware, including noise and limited qubit connectivity, which can cause errors and reduce the accuracy of the quantum algorithms.

Our study highlights the following key findings:

Accuracy: The Neural Network achieved the highest accuracy across all three datasets which can be attributed to the Neural Network's ability to capture complex patterns and relationships between the datapoints. This was followed by the SVM algorithm which demonstrated its effectiveness in separating the different classes in the datasets. However, VQC and QSVC had lower accuracy scores which indicates that there is room for improvement in the quantum algorithms implemented using Qiskit. These Quantum algorithms are still in their early stages, as Qiskit lacks a full release and is being constantly updated [31], and require more advanced techniques and optimizations to achieve higher accuracy comparable to classical algorithms.

CPU Usage: QSVC's higher CPU usage can be attributed to the computational demands of quantum circuit simulations and operations. Quantum algorithms involve complex calculations and transformations on qubits, which requires significant computational resources. In contrast, the Neural Network and SVM algorithms had relatively lower CPU usage, as they primarily involve comparatively simple matrix computations and optimization algorithms. VQC exhibited the lowest CPU usage among the quantum algorithms, indicating its efficiency in utilising quantum resources.

Memory Usage: All algorithms demonstrated relatively low memory usage throughout the study, with no significant differences observed between the classical and quantum algorithms. This observation can be attributed to the relatively small size of the datasets used in the experiments necessitated by the limitations on available qubit resources in currently accessible quantum devices. As a result, the memory requirements for both classical and quantum algorithms remained comparable.

Noise Levels: The lower noise levels observed in the simulated quantum algorithms can be attributed to the absence of physical noise and error sources present in real quantum devices. Simulated quantum algorithms run in noiseless environments, allowing for more accurate and reliable results. In contrast, the actual quantum devices exhibited higher levels of noise, which can arise from various factors such as imperfect qubit gates, environmental interactions, and limited qubit coherence times. The presence of noise in the quantum devices can lead to decreased accuracy and reliability in the outcomes of quantum computations, as observed in the lower accuracy scores

for the real quantum devices compared to the simulated ones.

These findings have important implications for the practical application of quantum machine learning algorithms. While the simulated results demonstrated the potential of quantum algorithms, the performance gap observed on real quantum hardware highlights the current limitations of currently available quantum hardware. Future advancements in quantum computing technology, such as improved qubit coherence and reduced noise levels, are crucial to utilising the full potential of quantum machine learning algorithms.

Chapter 8: Conclusions

8.1 Performance Comparison

The performance comparison between classical machine learning algorithms (Neural Network and SVM) and quantum machine learning algorithms (VQC and QSVC) revealed several important insights into the capabilities and limitations of the various approaches. While the classical algorithms demonstrated superior accuracy across all three datasets, the quantum algorithms showcased competitive performance in simulated environments. However, when deployed on real quantum devices, the quantum algorithms exhibited significant accuracy degradation, which we can primarily attribute due to the observed high noise levels and limited qubit connectivity.

Havlicek et al. [20] highlight the potential of quantum algorithms to excel in exponentially large feature spaces, utilising the abilities of quantum systems. However as shown by our results, current performance is hamstrung and constrained by limitations in available quantum hardware. The persistent challenge of noise in quantum computing introduces errors and constrains the accuracy of quantum algorithms. As observed in the real quantum devices utilised in this study, namely Quantinuum, Rigetti, and IonQ, the noise levels hindered the ability of the VQC and QSVC to achieve competitive accuracy, highlighting the need for advancements to be made in quantum hardware and noise mitigation.

Despite the limitations encountered on the real quantum devices, the simulated environments provided a valuable insight into the potential of quantum machine learning algorithms. The competitive accuracy scores achieved in the simulations demonstrate the capability of VQC and QSVC to perform well in noise-free and idealized quantum computing scenarios. These results prove the importance of advancing both quantum algorithms and quantum hardware to fully exploit the benefits of quantum machine learning.

8.2 Limitations

As discussed above and observed in this study, the limitations of current quantum hardware has a significant impact on the performance of quantum algorithms. Noise, arising from environmental interference and imperfections in qubit coherence, introduces errors during computation, thereby reducing the accuracy of the quantum algorithms. Moreover, the connectivity limitations between qubits on existing quantum devices restrict the ability to perform certain quantum operations efficiently, potentially hindering the performance of quantum algorithms.

Another contributing factor to the performance gap is the impact of decoherence. Quantum systems are highly sensitive to their environment, leading to the degradation of quantum states over time. Decoherence can disrupt the delicate superposition and entanglement states required for quantum algorithms, thereby reducing their effectiveness. As a result, mitigating decoherence effects is crucial for achieving higher accuracy in quantum machine learning.

As quantum hardware continues to evolve and improve, it has the potential to overcome the current limitations, enabling quantum algorithms to scale efficiently. Advancements in error correction

techniques, reduction of noise sources, and increased qubit connectivity will be vital in realizing the full potential of quantum machine learning. The results of this study emphasize the need for ongoing research and development efforts to address the challenges posed by noise and improve the performance of quantum algorithms on real quantum hardware.

In conclusion, while the classical machine learning algorithms showcased superior accuracy in this study, the quantum machine learning algorithms demonstrated potential in simulated environments. The current limitations of quantum hardware, particularly in terms of noise and qubit connectivity, present challenges for deploying quantum algorithms on real quantum devices. However, as advancements in quantum technology continue to progress, the future holds promise for the scalability and effectiveness of quantum machine learning algorithms.

8.3 Future Directions

To overcome the limitations of current quantum hardware and improve the performance of quantum machine learning algorithms, future research should focus on several key areas.

8.3.1 Error Mitigation and Error-Correction Techniques

Developing better error mitigation and error-correction techniques specific to quantum machine learning is integral to the future of Quantum Machine Learning. These techniques should mitigate the impact of noise and decoherence, effectively enhancing the accuracy and reliability of quantum algorithms. Various error-correction codes, such as the surface code, hold promise for reducing errors and increasing the fault-tolerance of quantum computations.

8.3.2 Hardware Advancements

Advancements in quantum hardware, such as increased qubit coherence times, reduced noise levels, and improved qubit connectivity, are vital for enhancing the performance of quantum machine learning algorithms. Continued research and engineering efforts in the field of quantum computing hardware will enable the execution of more complex and accurate computations.

8.3.3 Algorithmic Improvements

Further optimization of quantum machine learning algorithms is necessary to enhance their efficiency and accuracy. Researchers should explore techniques that leverage quantum algorithms' unique properties, such as quantum parallelism and quantum entanglement, to develop novel approaches for data preprocessing, feature selection, and model training.

8.3.4 Hybrid Approaches

Hybrid approaches as described in 3.5 that combine classical and quantum computing methods hold potential for bridging the gap between quantum and classical machine learning. By utilising

the strengths of both approaches, hybrid models can mitigate the impact of noise and enhance accuracy. Such approaches can also enable the utilisation of quantum algorithms as subroutines within larger classical machine learning and artificial intelligence pipelines.

8.4 Ethical Considerations

As quantum computing advances, it is important to consider the ethical implications of its integration with machine learning. Quantum machine learning has the potential to revolutionise fields such as healthcare, finance, and cybersecurity. However, it also raises concerns regarding data privacy, algorithm bias, and the potential for malicious uses.

The increased computational power of quantum systems may enable more accurate and sophisticated machine learning models, but it also has the potential to compromise sensitive information if not handled properly. Encryption techniques and privacy-preserving algorithms are essential to protect individuals' data and must be developed with quantum-resistant encryption techniques in order to safeguard them against potential future threats of quantum-based encryption breaches.

Additionally, algorithm bias is a persistent issue present in classical machine learning algorithms and may persist in quantum machine learning models. Bias detection and mitigation techniques should be developed specifically for quantum algorithms to ensure fairness and avoid perpetuating existing biases or introducing new ones.

To address these ethical considerations, interdisciplinary collaboration between researchers, industry professionals, and policy-makers is crucial. Guidelines and regulations should be established to govern the ethical use of quantum machine learning and to ensure its responsible application in society.

8.5 Barriers to Quantum Development

During the course of our study, we encountered certain barriers that highlighted the challenges in the field of quantum development. One such challenge was the deployment process. We initially intended to deploy our algorithms to IBM Quantum for testing purposes. However, we faced difficulties in the deployment process, and ultimately found it more convenient to deploy to Azure Quantum instead. This experience suggests that the ease of deployment can vary across different quantum computing platforms, and choosing the right platform can significantly impact the development workflow.

Additionally, in some instances, the documentation provided by IBM for their Qiskit quantum development framework was found to be incomplete or vague, which can pose difficulties for newcomers to the field. While it is understandable that quantum computing is still in its early stages of development [31], accessible, comprehensive, and understandable documentation is crucial in supporting the learning and adoption of quantum programming frameworks. Improvements in documentation quality and accessibility would greatly benefit the quantum development community by enabling a more forgiving learning curve and reducing barriers to entry.

These barriers emphasise the importance of user-friendly tools, robust documentation, and streamlined deployment processes in the quantum development ecosystem. Among the other results, our

study illustrates the hurdles presented by these challenges. Overcoming them by providing accessible resources will play a pivotal role in attracting new talent, fostering innovation, and advancing the field of quantum computing.

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